FlightSim + HHMARL_3D_NEW

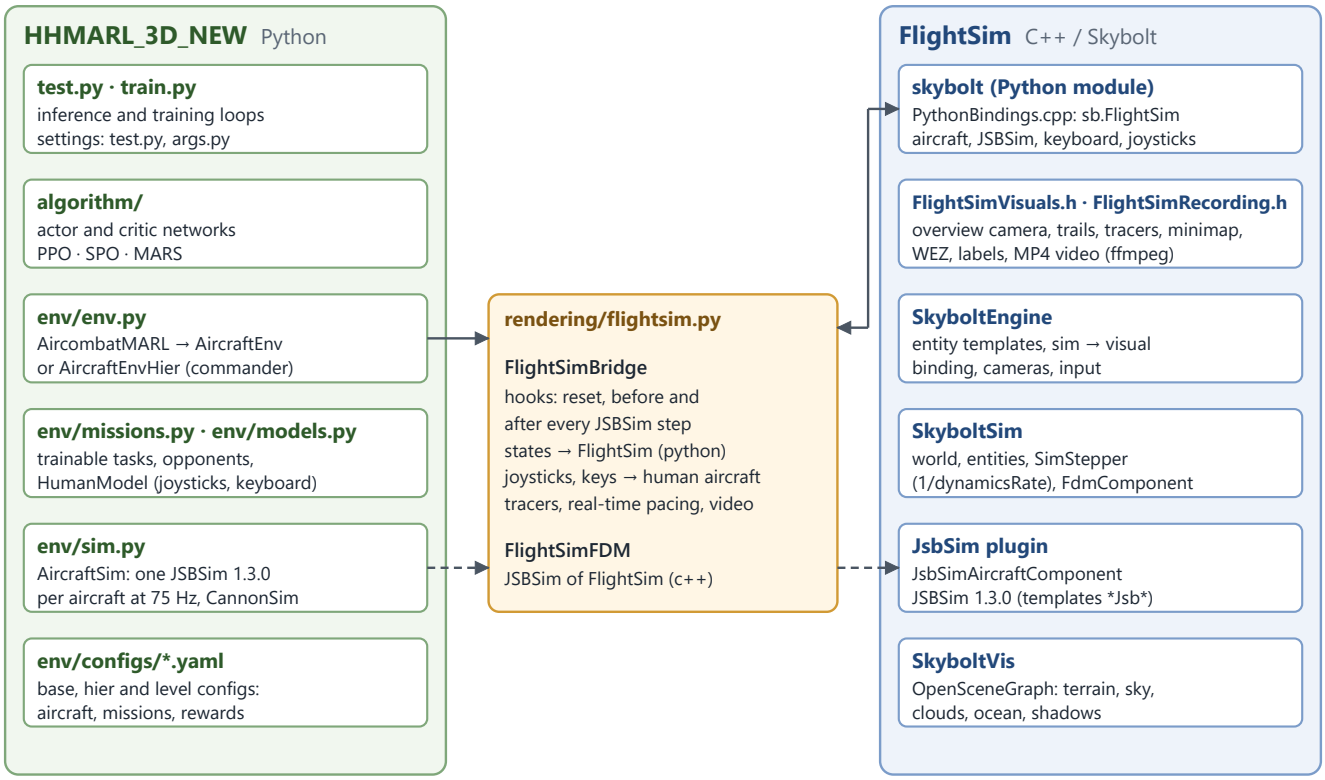
Components, architecture and parameters

Status 30 October 2026 · FlightSim Python API version 10 · JSBSim 1.3.0 at 75 Hz

1 Overview

- **HHMARL_3D_NEW**: multi-agent air combat environment in Python on JSBSim, with its own on-policy MARL training (MAPPO, MASPO, MARS). Training always runs here, without FlightSim.
- **FlightSim**: a modified Skybolt (C++, OpenSceneGraph) with the Python module `skybolt`. It is used for inference only: realistic rendering, video recording, optionally the C++ JSBSim, and a human pilot with joysticks or the keyboard.
- The repositories are separate clones side by side (`Projects/FlightSim`, `Projects/HHMARL_3D_NEW`). HHMARL imports FlightSim only when it is used.
- The HHMARL env always computes observations, rewards and terminations. FlightSim displays the aircraft, reads the joysticks and the keyboard or simulates the flight dynamics.

2 Architecture



Solid arrows: always used with FlightSim. Dashed: only with `JSBSIM_STATES = "c++"`, where AircraftSim calls the JSBSim of FlightSim.

One env step with FlightSim

1. The policies act. Each mission turns its action into JSBSim flight controls (aileron, elevator, rudder, throttle) and fires the cannon.
2. `sim_steps` (10) JSBSim steps of 1/`sim_freq` s (1/75 s) follow. Before each step, FlightSimBridge hides aircraft that were shot down or crashed, writes the controls of the human (joysticks or keyboard) and, in `c++` mode, advances the JSBSim of FlightSim (`step(1)`). After each step, it copies the states of the Python JSBSim into FlightSim (python mode, `step(1)`), shows tracers while a cannon fires and renders in real time (`speed`). While recording, every frame of the video is rendered, even if the simulation then runs slower than real time.
3. The env computes observations, rewards and terminations, as without FlightSim.

Commander (`hier`): one env step assigns a low-level policy (cannon, track or escape) to each aircraft and runs it for `hl_sim_steps` low-level steps. With FlightSim, all aircraft advance in lockstep (all set their controls, then all step together).

JSBSim modes

JSBSIM_STATES	Flight dynamics	FlightSim templates
`"python"` (default)	jsbsim Python package of HHMARL; FlightSim only displays the states	`SepecatJaguarBlue`, `SepecatJaguarRed`
`"c++"`	JSBSim plugin of FlightSim; `FlightSimFDM` replaces `jsbsim.FGFDMExec` in AircraftSim	`SepecatJaguarJsbBlue`, `SepecatJaguarJsbRed` (JSBSim model `f16` only)

Both modes use JSBSim 1.3.0 at 75 Hz and give identical trajectories; `Tools/Compare_JSBSim_Python_Cpp.py` checks this.

3 FlightSim

3.1 Components

Component	Where	Role
skybolt Python module	`src/Skybolt/SkyboltPythonBindings/PythonBindings.cpp`	`sb.FlightSim`: window, aircraft, JSBSim access, cameras, keyboard, labels
Joysticks	`src/Skybolt/SkyboltPythonBindings/FlightSimJoysticks.h`	joysticks, throttles and pedals of the manual aircraft (`Tools/joysticks.json`), read with SDL 2.32.10
Visuals	`src/Skybolt/SkyboltPythonBindings/FlightSimVisuals.h`	overview camera (F3), trails, cannon tracers, minimap (key M), weapon engagement zones (key Z)
Recording	`src/Skybolt/SkyboltPythonBindings/FlightSimRecording.h`	MP4 video of the window view (key R), encoded by ffmpeg in a background thread
SkyboltSim	`src/Skybolt/SkyboltSim`	world, entities and components; `SimStepper` (fixed steps of 1/dynamicsRate); `FdmComponent` (flight dynamics interface)
JSBSim plugin	`src/Skybolt/SkyboltEnginePlugins/JsbSim`	JSBSim 1.3.0 for templates with a `jsbSimAircraft` component
SkyboltEngine	`src/Skybolt/SkyboltEngine`	engine root, entity templates, sim-to-visual binding, camera controllers, keyboard and mouse
SkyboltVis	`src/Skybolt/SkyboltVis`	OpenSceneGraph rendering: planet, terrain, sky, clouds, ocean, shadows
Assets	`Assets/`	templates (`Movers/Entities`), models, F-16 of the manual aircraft (`Movers/JSBSimManual`), shaders (`Core`), planet and map tiles (`Globe`, `mapbox_*_ch`)
Dependencies	`conanfile.py`, `Conan/Recipes`	Conan 2 packages, incl. the JSBSim 1.3.0 recipe and SDL 2.32.10 (joysticks)
Path setup	`flightsim_env.py`	`import flightsim_env` before `import skybolt`: finds the build, sets library and asset paths and `FLIGHTSIM_FFMPEG`, reads `Tools/mapbox_token.txt`, sets `FLIGHTSIM_JOYSTICKS`, selects the dedicated GPU (Windows: graphics preference of python.exe; WSL2, Linux: environment variables of Mesa and NVIDIA)
Examples	`Test_FlightSim_Windows.py` `Test_FlightSim_Linux.py`	manual flight (joysticks or keyboard) of a JSBSim aircraft next to an aircraft moved from Python

3.2 Python API: sb.FlightSim

Constructor argument	Default	Meaning
`x, y, width, height`	`0, 0, 1080, 720`	window position and size in pixels
`manual`	`False`	the joysticks (3.3) or the keyboard fly the aircraft of `set_manual_aircraft(id)`
`multiwindow`	`False`	second window with a free camera
`dynamicsRate`	`75.0`	simulation steps per second
`headless`	`False`	no window, e.g. for comparisons (not with manual or multiwindow)

Area	Calls and properties
Aircraft	`add_aircraft(id, template, position, rpy, velocityNed)` · `set_aircraft_state(id, position, rpy, velocityNed=None)` · `remove_aircraft(id)` · `set_aircraft_visible(id, visible)` · `set_dynamics_enabled(id, enabled)` · `get_aircraft_ids()` · `get_aircraft(id)`

Area	Calls and properties
Units	`position`: `sb.LatLonAlt(lat, lon, alt)`, geodetic radians, m above sea level · `rpy`: `sb.Vector3` roll, pitch, yaw in radians (north-east-down) · `velocityNed`: north, east, down in m/s
JSBSim	`get_property(id, name)` · `get_properties(id, names)` · `set_property(id, name, value)` · `query_property_catalog(id)` · `run_ic(id)` · `get_num_engines(id)` · `init_engine_running(id, engine)` · `get_steady_state(id)` · `set_external_control(id, enabled)`: flight controls only from the properties, not from the joysticks or the keyboard
Simulation	`step(substeps=1)` · `render()`: `False` once the window is closed or Esc is pressed · `frame(dt)`: steps for `dt` seconds and renders · `sim_time` · `dynamics_step_size`
Window	`vsync` (key V, on: render() waits for the refresh of the monitor) · `fullscreen` (key B; a change of the window size stops a recording)
Camera	`select_camera_view(view)` with `"Cockpit"`, `"Follow"`, `"Overview"` or `"Free"` · `set_camera_target(id)`
Labels, trails	`set_aircraft_label(id, text)` · `set_aircraft_color(id, (r, g, b))` (linear RGB) · `labels_visible` · `trails_visible` · `trail_length` (m, 30000) · `clear_trails()` (also the paths on the minimap) · `set_aircraft_throttle(id, throttle)`: nozzles of the camera target, without JSBSim
Minimap	`minimap_visible` (key M, on) · `minimap_range` (m, 20000: outer ring, farther aircraft at the ring with their distance) · `minimap_heading_up` (key N; default: with `manual`): map of the aircraft around the camera target in the bottom right corner, arrows in their colors with their labels and distances (3D) to the camera target, the paths of the other aircraft in the last 30 s as dots; in the free view (F4) the camera and its field of view
WEZ	`set_aircraft_team(id, team)`: aircraft of different teams are opponents (none without a team) · `wez_visible` (key Z, on) · `wez_range` (m, 3500) · `wez_angle` (degrees, 10: half angle of the cone around the velocity): in the cockpit and follow views (F1, F2), a light green circle ahead of the camera target while an opponent is in its weapon engagement zone, and a red beam like a flashlight from each opponent that has it in its zone, most opaque at the opponent (or at the edge of the window in its direction if it is out of view) and fading out towards the camera target (a red border around the window for an opponent nearly behind the camera)
Cannon	`fire_tracer(id, speed=1000.0, lifetime=1.0)`: dark tracer line · `clear_tracers()`
Recording	`start_recording(path=None)`: returns the video file · `stop_recording()` · `recording` · `recording_frame_due` · `recording_directory` (`"recordings"`) · `recording_fps` (30), see 3.6
Manual flight	`set_manual_aircraft(id)` · `get_manual_controls()`: aileron, elevator, rudder, throttle · `consume_fire_requests()`: presses of the fire button or Space since the last call · `fire_pressed`: held
Low level	`engineRoot` · `world` · `visRoot` · `window` · `camera` · `sb.captureScreenshot(app.visRoot)`

3.3 Keyboard, mouse and joysticks

Key	Action
F1 · F2 · F3 · F4	cockpit · follow (external) · overview of all aircraft · free camera
Tab	next aircraft as camera target
L · T · R	labels · trails · start/stop recording a video
M · N	minimap · minimap north-up/heading-up (shown at the top of the window)
Z	weapon engagement zones in the cockpit and follow views on/off (shown at the top of the window)
V · B	vsync on/off (shown at the top of the window and in the help) · full screen on/off
H · Esc	help · exit
Mouse, wheel	rotate the camera; in the overview, rotate around the aircraft and zoom with the wheel
A/D · W/S	overview: rotate · zoom (unless they fly the manual aircraft); free camera: move at 2 km/s (Shift held: 10 times faster, Ctrl held: 10 times slower)
Manual flight	without joysticks: Up/Down pitch down/up · Left/Right roll · A/D rudder · W/S throttle · Space cannon
Joysticks	the devices of `Tools/joysticks.json` fly the manual aircraft while they are connected, each control otherwise with its keys: stick (roll, pitch), throttle lever, rudder pedals, trigger (fires while held). Device names and axis and button numbers as shown by `Tools/joysticks.py`, optionally `invert` and `deadzone`; read when FlightSim starts, so changes need no rebuild

3.4 Setup, tools and settings

Item	Parameters and use
`setup_windows.bat`	full setup; `-BuildDir <folder>`, `-PythonExe <python.exe>`
`setup_linux.sh`	full setup; `--build-dir <folder>`, `--python <python3.11>`, `--jobs N`
`Tools/update_pythonbindings_windows.bat` `Tools/update_pythonbindings_linux.sh`	rebuild after C++ changes (Linux: `--jobs N`). Run the full setup instead after adding or removing source files or changing dependencies.
`Tools/Compare_JSBSim_Python_Cpp.py`	flies an F-16 in both JSBSims for 120 s at 75 Hz (`SECONDS`, `RATE`) and prints the differences
`Tools/Tile_Map_Downloader.py`	downloads (or unzips from `Assets`) the offline map tiles of Switzerland; `--zoom 15 16` for more detail (needs a Mapbox token); area: `CENTER_LATITUDE`, `CENTER_LONGITUDE` in the script
`Tools/mapbox_token.txt`	optional Mapbox access token (Git stores only the comments, filter set up by the setup). Without it, only the offline map tiles are shown; with it, the missing tiles are loaded from Mapbox
`Tools/joysticks.json`	device and axis or button of each flight control of the manual aircraft: roll, pitch, throttle, rudder, fire (3.3)
`Tools/joysticks.py`	shows the names and the axes, buttons and hats of the connected joysticks, throttles and pedals
`Tools/Scale_OSG_Files.py`	scales a model, e.g. by `8.0` for the enlarged *Big models; `--to-osgt` converts it to text
`Tools/build_instructions_pdf.py`	builds this `instructions.pdf` into the current folder (Windows fonts)
Packages of .venv	installed by the setup: `conan` 2.22.0, `jsbsim` 1.3.0, `requests`, `imageio-ffmpeg` 0.6.0 (ffmpeg of the videos), `reportlab` 5.1.0 (this PDF), `pygame-ce` 2.5.8 (`Tools/joysticks.py`)
Environment variables	`FLIGHTSIM_BUILD_DIR`: build folder (default `../FlightSimBuild`) · `FLIGHTSIM_DIR`: FlightSim folder for HHMARL (default `../FlightSim`) · `FLIGHTSIM_FFMPEG`: ffmpeg of the videos (default: the one of imageio-ffmpeg, otherwise ffmpeg on PATH) · `FLIGHTSIM_MAPBOX_TOKEN`: Mapbox token (default: the one of `Tools/mapbox_token.txt`) · `FLIGHTSIM_JOYSTICKS`: joystick mapping (default `Tools/joysticks.json`, empty: keyboard only)
`Settings.json`	Windows `%LOCALAPPDATA%\Skybolt`, Linux `~/Skybolt`: `display` (`multiSampleCount`, `vsync`, `texturePoolSizeMB`), `shadows`, `clouds`, `mouse.sensitivity`, `tileApiKeys` (keep empty, the Mapbox token is in `Tools/mapbox_token.txt`), `jsb.rootDir` (JSBSim aircraft data)
`PlanetEarth.json`	`Assets/Globe/Entities`: `maxLevel` of the map tiles; raise it after downloading more zoom levels; `fallback`: online Mapbox tiles, used with a token
Templates	`Assets/Movers/Entities`: `SepecatJaguarBlue/Red` (moved from Python), `SepecatJaguarJsbBlue/Red` (JSBSim `f16`), `SepecatJaguarJsbBlue/RedManual` (F-16 with corrected rudder for the manual aircraft, `aircraftDir`), `SepecatJaguarBlueBig/RedBig` (enlarged models)

3.5 Visual constants (C++)

Element	Values in `FlightSimVisuals.h` (rebuild after changing)
Overview camera	vertical field of view 0.6 rad, looks down at 12°, margin 1.15 around the aircraft, at least 60 m away, zoom factor up to e^3
Trails	a point every 25 m, about 3 px wide, opacity 0.9 fading quadratically along `trail_length`
Tracers	20 m dark line, about 4 px wide, starting 10 m ahead of the aircraft
Nozzles	discs at the nozzle exits of the camera target (`nozzles` of the entity template), linearly from black at idle to dark red (linear RGB 0.045, 0.0012, 0) at full throttle (`fcs/throttle-cmd-norm`)
Minimap	`TacticalMinimap`: black background (opacity 0.7) of 28 % of the window height, 2 % from the bottom right corner; outer ring 25 % of the window height wide; arrows 2 % long (camera target 2.4 %), texts 1.6 % in the aircraft colors mixed 45 % with white, beside the arrows where they cover the least of the other symbols and texts; distances in steps of 10 m below 1 km, 0.1 km below 10 km; paths: a point every 0.5 s, dots 0.35 % large every 0.75 % of the window height along them; free camera: light gray camera 2.25 % long with dashed lines along its horizontal field of view to the outer ring

Element	Values in `FlightSimVisuals.h` (rebuild after changing)
WEZ	`WezIndicators`: green circle (linear RGB 0.3, 0.9, 0.3) with a radius of 7.2 % of the window height, opacity 0.2 inside and 0.5 on its rim, ahead of the nose at the distance of the nearest opponent in the zone; red beams (linear RGB 1, 0.03, 0.02) like flashlights from the opponent on the screen (or the edge of the window in its direction if it is out of view) to the camera target (the center in the cockpit): opacity 0.5 at the source, falling with the power 1.5 to 0 at the camera target; half opacity 1.5 % of the window height from the axis at the source, widening at 12° to each side, with slightly sharp sides (sharpness 1.5); within 30° of the direction behind the camera they turn gradually into a red border around the window (opacity 0.5 at the edge); drawn below the labels and the minimap

3.6 Video recording

Aspect	Behavior
Start, stop	key R or `start_recording()` / `stop_recording()`; also stops when the window is closed or the `FlightSim` is deleted; a red dot in the top left corner shows the recording
Content	the view of the window (selected camera with labels, trails, minimap, WEZ and cockpit HUD if shown), without the help message and the red dot; size of the window (rounded down to even pixels)
File	`<recording_directory>/flightsim_<date>_<time>.mp4`, or the path given to `start_recording(path)`; a relative folder is relative to the current directory
Frame rate	`recording_fps` (30) frames per second of simulation time, each from the simulation step closest to its time, so the video plays at simulation speed (at 75 Hz alternately 2 and 3 steps apart)
Skipped rendering	programs that skip frames to keep up with real time must still render when `recording_frame_due` is `True`, otherwise frames repeat (FlightSimBridge does)
Encoding	H.264 (libx264, CRF 18, yuv420p, BT.709) by the ffmpeg of `FLIGHTSIM_FFMPEG`; not with `headless`

4 HHMARL_3D_NEW

4.1 Components

Module	Role
`train.py`	training; `process_args` applies experiment-specific overrides
`test.py`	inference over many episodes (win and loss rates), with rendering
`args.py`	all arguments of training and inference (command-line flags)
`algorithm/`	learner (PPO, SPO, MARS), actor and critic networks, value normalization
`env/env.py`	`AircombatMARL` (interface of the learner) → `AircraftEnv` (low level) or `AircraftEnvHier` (commander)
`env/sim.py`	`AircraftSim` (one JSBSim per aircraft) and `CannonSim`
`env/missions.py`	trainable tasks: track, cannon, escape, hier
`env/models.py`	scripted and pre-trained opponents, `HumanModel` (joysticks or keyboard via FlightSim)
`env/rewards.py`, `env/terminations.py`	reward terms and episode ends
`env/configs/`	`base.yaml`, `hier.yaml`, `levels/<task>_L<n>.yaml` (deep-merged)
`env/base_actor/`, `env/catalog.py`	scripted base actor and JSBSim property catalog from LAG
`rendering/`	`render.py`, `scene.py`: PyGame renderer · `flightsim.py`: FlightSim bridge
`runs/`	checkpoints and TensorBoard logs: `runs/<task>_<model>_<run_name>`
`recordings/`	videos recorded with key R in the FlightSim window (ignored by git)

4.2 Inference settings (test.py)

Setting	Default	Meaning
RENDER	True	render while testing
RENDERER	"flightsim"	"pygame" (simple) or "flightsim"
RESTORE_PATH	runs\cannon_L4_f16	checkpoint file, or run folder (its latest checkpoint)
TASK, LEVEL	"cannon", 4	task and level of the policy; commander: --hierarchy_level 2
INIT_STATES	None	per aircraft, relative to the battle field center: north/east in m (or lat/lon in degrees), alt in m, speed in m/s, heading, roll, pitch in degrees; None: config
JSBSIM_STATES	"python"	"python" or "c++", see section 2
MANUAL, MANUAL_AGENT	False, "B2"	a human flies MANUAL_AGENT with the joysticks or the keyboard: a copy of B1 if the config has no such aircraft (track, cannon, escape), otherwise it replaces the RL agent (hier: B3)
LOCATION	(8.965318, 45.975305)	battle field center (longitude, latitude), here Lugano
MINIMAP, MINIMAP_RANGE	True, 20000	minimap of the aircraft around the camera target (3.2), and its range in m
WEZ	True	weapon engagement zones of the cannons in the cockpit and follow views (3.2)
run_inference(num_episodes)	500	number of episodes

4.3 FlightSim bridge (env config "flightsim")

Options of FlightSimBridge in rendering/flightsim.py. test.py passes jsbsim_states, manual_agent, minimap, minimap_range and wez; add further options to kwargs["flightsim"] there.

Option	Default	Meaning
jsbsim_states	"python"	"python" or "c++"
manual_agent	None	ID of the aircraft flown by the human, e.g. "B2"
window	(10, 30, 1600, 900)	x, y, width, height of the window
speed	1.0	simulated seconds per real second; None: as fast as possible
camera_view	None	"Cockpit", "Follow", "Overview" or "Free"; None: cockpit with a human, otherwise overview
trail_length	30000.0	length of the trails in m
templates	None	templates per team, e.g. {"R": "SepecatJaguarRedBig"}; with c++, they need a jsbSimAircraft component
tracer_interval	5	JSBSim steps between tracers while a cannon fires
headless	False	no window (not with manual_agent)
recording_dir	None	folder of the videos of key R; None: recordings/ of HHMARL_3D_NEW. While recording, every video frame is rendered (no frames are skipped), so a slow PC runs slower than real time
minimap	True	minimap of the aircraft around the camera target (key M)
minimap_range	20000.0	radius of the minimap in m
minimap_heading_up	None	True: heading-up, False: north-up (key N); None: heading-up with a human, otherwise north-up
wez	True	weapon engagement zones (key Z) of the cannons (CannonSim: Rmax_wez, omega_wez), with the teams of the env
TEAM_COLORS	B (0.05, 0.3, 1.0) R (1.0, 0.06, 0.04)	module constant: colors of the labels, trails and minimap per team (linear RGB)
FLIGHTSIM_API_VERSION	10	module constant: minimum version of the FlightSim build (10: weapon engagement zones)

4.4 Environment (AircombatMARL(task, **kwargs))

Argument	Values	Meaning
`task`	`"track/L4"`, `"cannon/L1"`, `"escape/L2"`, `"hier"`	config: `base.yaml` + level file or `hier.yaml`
`hierarchy_level`	`1` or `2`	1: low-level policies, 2: commander
`training_level`	integer	stage of the random initial states (set by train.py and test.py)
`rendering`, `render_view`, `render_samples`	`False`, `0`, `4`	PyGame rendering, viewpoint, anti-aliasing samples
`flightsim`	`None`	options of 4.3; `None`: without FlightSim
`config_overrides`	`None`	deep-merged into the config, e.g. `{"battle_field_center": [lon, lat, 0.0]}`
`reset(seed, options)`	`{"init_states": ...}`	initial states as `INIT_STATES`

4.5 YAML configuration (env/configs)

Key	Default	Meaning
`sim_freq`	`75`	JSBSim steps per second
`sim_steps`	`10`	JSBSim steps per env step (0.133 s)
`hl_sim_steps`	`10` (hier)	low-level steps per commander step
`horizon`	`600`; hier `100`	env steps per episode
`map_size`	`50`; hier `80`	arena size in km; farther than map_size/2 from the center is a crash
`altitude_limit`, `altitude_limit_max`	`2500`, `10000`	allowed altitudes in m
`acceleration_limit_x/_y/_z`	`9.0`	overload limit in g
`rnd_init`	`true`	random initial heading, position, altitude and speed
`battle_field_center`	`[0.0, 0.0, 0.0]`	longitude, latitude, altitude (test.py: `LOCATION`)
`num_rl_agents`, `num_opp_agents`	hier `3`, `3`	team sizes of the commander config
`opp_hier_strategy`	`mixed`	commander opponents: `mixed` (cannon/track/escape 40/40/20 %), `aggressive` (70/20/10 %), `random`
`sp_policy_name`	L3 levels	policy of the self-play opponent, e.g. `track_L2_f16`
`aircraft.<ID>`	`B1`, `R2`, …	first letter = team; `model` (`f16` or `A4`), `mission` (4.6), `num_missiles` (cannon shots, 10), `init_state` (`ic_long_gc_deg`, `ic_lat_geod_deg`, `ic_h_sl_ft`, `ic_psi_true_deg`, `ic_u_fps`), `unlimited_fuel` (`true`)
`<Reward>_scale`, `<Reward>_potential`	`1.0`, `false`	per reward term: `EventDrivenReward`, `ShootReward`, `ShootPenaltyReward`, `HeadingReward`, `DistanceReward`, `FleeReward`; also `ShootReward_orientation_kill`, `ShootPenaltyReward_posture_punish`, `HeadingReward_track`, `HeadingReward_opposite`

4.6 Missions and interfaces

Mission (`aircraft.<ID>.mission`)	Behavior
`task_track`, `task_cannon`, `task_escape`	trainable low-level skills: pursue, attack with the cannon, evade
`task_hier`	trainable commander: picks `cannon_L4`, `track_L4` or `escape_L2` per aircraft
`model_L1Maneuver`, `model_L2Maneuver`	scripted opponents (L1, L2): random turns and shots · precise tracking and shooting; based on the LAG base actor (`model_track`)
`model_SPManeuver`	self-play opponent (L3) with the pre-trained policy `sp_policy_name`
`model_strategy`	L4 opponent: mixed strategy over the pre-trained L3 (escape: L1) policies
`model_hier`	commander opponent with `opp_hier_strategy`
`model_human`	human pilot with joysticks or the keyboard via FlightSim (`MANUAL` in test.py); its F-16 has the corrected rudder of FlightSim's `Assets/Movers/JSBSimManual`

Interface	Definition
Low-level action	`MultiDiscrete([17, 11, 9, 9, 2])`: aileron, elevator, rudder, throttle, shoot
Low-level observation	30 values in [-5, 5]: own position, velocity, body rates, attitude, cannon state and ammunition; closest opponent: relative position, velocity and angles
Commander action	`Discrete(3)`: cannon, track, escape
Cannon	range 3500 m, cone ±10°, hit probability 0.8, shots = `num_missiles`; RL agents: a shot fires until it hits or for 12 s, the next one after 16 s (`launch()`); human: fires while the button is held, unlimited ammunition (`fire()`)

4.7 Training arguments (args.py)

Group	Arguments and defaults, set as flags, e.g. `python train.py --task track --level 2`
Run	`task track`, `level 1`, `hierarchy_level 1`, `run_name ""`, `restore_path None`, `seed 1`, `total_timesteps 5000000`
Algorithm	`algorithm ppo` (ppo, spo, mars), `trust_region_clip 0.25`, `gamma 0.99`, `td_lambda 0.92`, `use_gae True`, `entropy_coef 0.01`, `entropy_decay None`
Networks	`shared_policies True`, `centralized_critic True`, `actor_hidden_dim 128`, `critic_hidden_dim 128`, `actor_num_layers 1`, `critic_num_layers 1`, `use_ReLU True`, `normalize_input True`, `init_orthogonal True`
Recurrence, attention	`rnn False`, `rnn_num_layers 1`, `data_chunk_length 10`, `attention False`, `attention_heads 2` (rnn and attention exclude each other)
Optimization	`optimizer Adam`, `learning_rate_actor 3e-4`, `learning_rate_critic 3e-4`, `lr_decay None`, `train_epochs 10`, `batch_size 10000`, `minibatch_size 5000`, `clip_gradients 10`
Losses	`value_loss_coef 1.0`, `clip_value_loss 1.0`, `huber_coef 10.0`, `normalize_advantage True`, `normalize_value True`, `normalize_popart False`, `policy_mask True`, `value_mask True`
Workers, hardware	`num_envs 20`, `cuda True`, `cuda_id 0`, `torch_deterministic False`, `torch_compile False`, `avail_actions False`, `log_every_step 100`, `render_samples 4`

5 Quick start

Goal	Command
Set up FlightSim	`setup_windows.bat` · `./setup_linux.sh`
Try FlightSim	`.venv\Scripts\python Test_FlightSim_Windows.py` · `.venv/bin/python Test_FlightSim_Linux.py`
After C++ changes	`Tools\update_pythonbindings_windows.bat` · `./Tools/update_pythonbindings_linux.sh`
Train low level	`python train.py --task track --level 1`, then `--level 2 --restore_path runs/track_L1_f16_<run_name>`
Train commander	`python train.py --task hier --hierarchy_level 2`
Inference	set the settings in `test.py`, then `python test.py` (commander: `RESTORE_PATH` of a hier run and `--hierarchy_level 2`)
Record a video	press R in the FlightSim window to start and again to stop (or `app.start_recording()`); MP4 files in `recordings/`
Update this PDF	`.venv\Scripts\python Tools\build_instructions_pdf.py`

The venv of HHMARL_3D_NEW needs Python 3.11, like the `skybolt` module of FlightSim, and `imageio-ffmpeg` of its `requirements.txt` for the videos.